Satya Teja Chukka

India   satyateja671@gmail.com   +91 6301304206
LinkedIn   GitHub   Portfolio

Technical Skills

- **Languages:** Python, JavaScript, C++, C, SQL
- **Frameworks & Libraries:** FastAPI, React.js, SQLAlchemy, Pydantic, Scikit-learn, TensorFlow, Keras, OpenCV, Pandas, NumPy
- **DevOps:** Docker, Git, GitHub
- **Databases:** PostgreSQL, MySQL, SQLite
- **Tools:** VS Code, PyCharm, Jupyter Notebook, Google Colab

Education

Gayatri Vidya Parishad College of Engineering (Autonomous) Expected 2027
B.Tech in Computer Science & Engineering (AI & ML)
- CGPA: 8.62/10
- **Relevant Coursework:** Data Structures & Algorithms, DBMS, OOP, Operating Systems, Computer Networks, Machine Learning, Deep Learning

Experience

Machine Learning Intern | Crescout.ai Remote | Mar 2026 – May 2026
- Built automated Kaggle execution workflows by dynamically injecting generated match IDs into notebooks and triggering remote video-processing pipelines.
- Integrated Kaggle and Google Drive workflows for video retrieval, preprocessing, and processed video storage automation.
- Worked on LSTM and dataset preparation for BiLSTM + Attention models for badminton shot analysis, next-shot prediction, rally classification, and winning probability estimation.
- Trained ResNet and MobileNet-based computer vision models for badminton court detection using masking-based learning pipelines and 200+ manually annotated gameplay frames, achieving 99.66% pixel accuracy with average inference times of 29.01 ms, 35.31 ms, and 59.78 ms per frame for 128x128, 192x192, and 256x256 resolutions respectively.

Projects

Wealth Sync | Personal Finance Web Application Live GitHub
- Built a full-stack finance platform with FastAPI, React, and PostgreSQL for transaction tracking, spending categorization, and dashboard analytics.
- Designed and implemented REST APIs using FastAPI, Pydantic, and SQLAlchemy for authentication, transaction processing, and reporting workflows.
- Architected PostgreSQL schemas and Alembic migration pipelines for scalable financial data storage and querying.
- Containerized backend and frontend services using Docker and docker-compose for scalable deployment workflows.

TubeRAG | YouTube RAG Platform GitHub
- Built a Retrieval-Augmented Generation (RAG) platform that enables natural-language question answering over YouTube channels, playlists, and videos with timestamped source citations.
- Designed a hybrid retrieval pipeline combining ChromaDB vector search, SQLite FTS5 keyword search, and Reciprocal Rank Fusion (RRF) to improve retrieval accuracy and relevance.
- Implemented parent-child chunk retrieval and optional FlashRank re-ranking to provide richer context and improve answer quality from video transcripts.
- Developed asynchronous multi-video ingestion workflows using FastAPI and asyncio, supporting concurrent transcript extraction, embedding generation, and indexing.
- Integrated local embeddings, BYOK API-key management, Server-Sent Events (SSE) streaming, and Dockerized deployment for a production-style AI application.

Certifications

- IBM (edX): Python Basics for Data Science, Analyzing Data with Python
- Coursera : Supervised Machine Learning: Regression and Classification, Advanced Learning Algorithms

Achievements

- Solved 570+ DSA problems across LeetCode and GeeksforGeeks.
- Achieved a highest LeetCode contest rating of 1801.
- Achieved 2 star rating on CodeChef.